

QuackBlocks — Palmer Penguins Report

 Live analytics inside Obsidian

Every table and chart in this document is generated by DuckDB WASM from SQL queries against local Parquet files. No external servers, no copy-pasting screenshots — just write SQL and hit export to PDF.

Executive Summary

 About the dataset

The Palmer Penguins dataset contains morphological measurements from three penguin species observed near the Palmer Station in Antarctica. It is a modern, drop-in replacement for the classic Iris dataset — with the added benefit of being adorable.

total_penguins	species_count	island_count	avg_mass_g	avg_flipper_mm
60	3	3	4147.2	200.9

Dataset overview

 PDF export

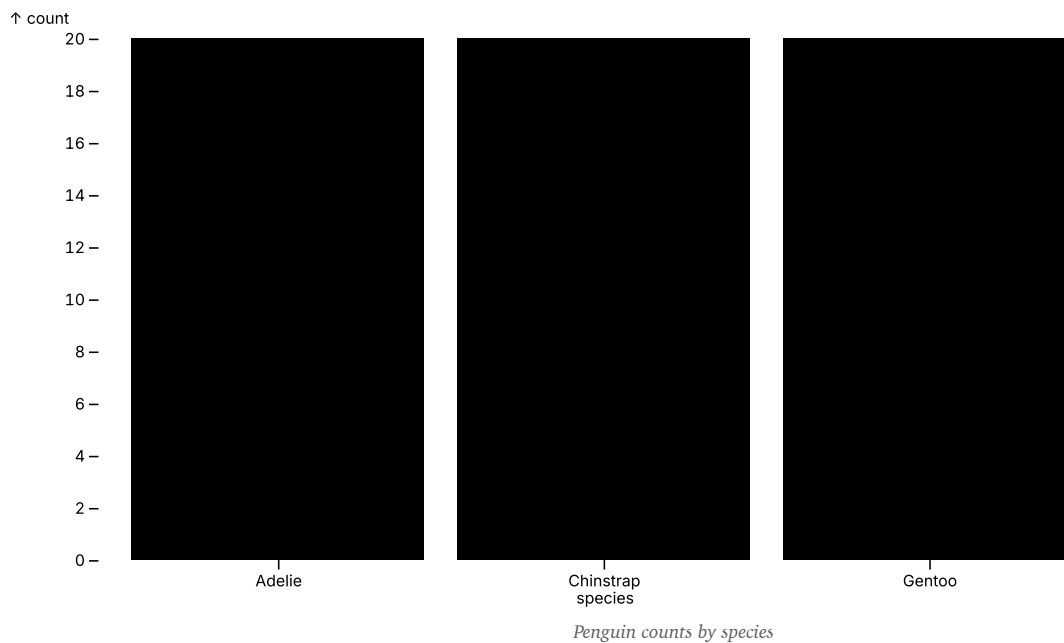
Try **File** → **Export to PDF** right now. Every chart below renders as SVG with explicit dimensions, so the exported PDF looks identical to what you see in Obsidian.

1. Species Overview

 Three species, three islands

Adelie penguins are found on all three islands, while Chinstrap and Gentoo each occupy a single island. This geographic isolation is a natural experiment in speciation.

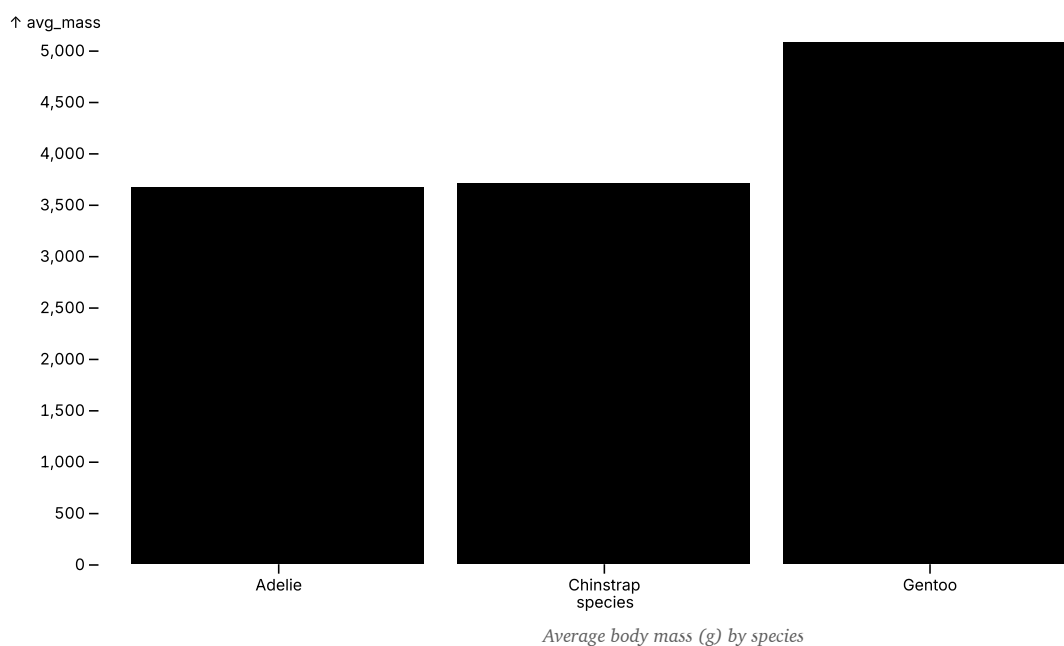
1.1 Population by species



1.2 Average measurements by species

🔍 Hover over bars

Tooltips are enabled by default. In the rendered view you can hover to see exact values.



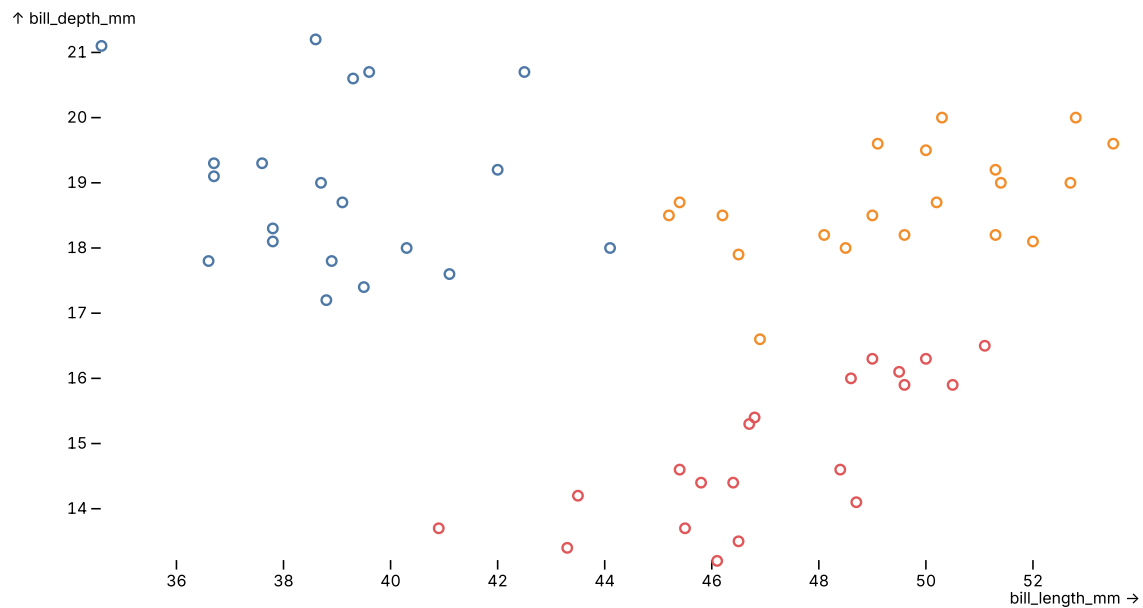
2. Morphology Analysis

📄 Bill length vs. bill depth

These two measurements are the classic diagnostic features for distinguishing penguin species. Adelie and Chinstrap cluster by bill depth, while Gentoo separates cleanly by length.

2.1 Bill dimensions scatter

■ Adelie ■ Chinstrap ■ Gentoo

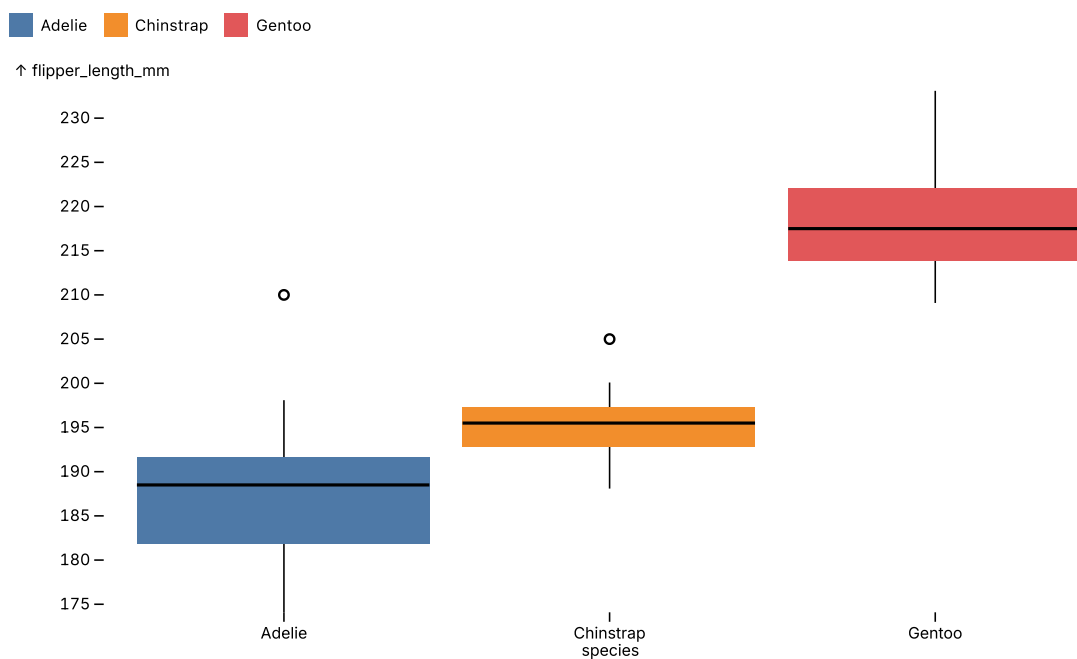


Bill length vs. bill depth, coloured by species

Notice the clustering

The three species form distinct clusters. This is why bill dimensions alone are enough for a highly accurate classifier.

2.2 Flipper length distribution



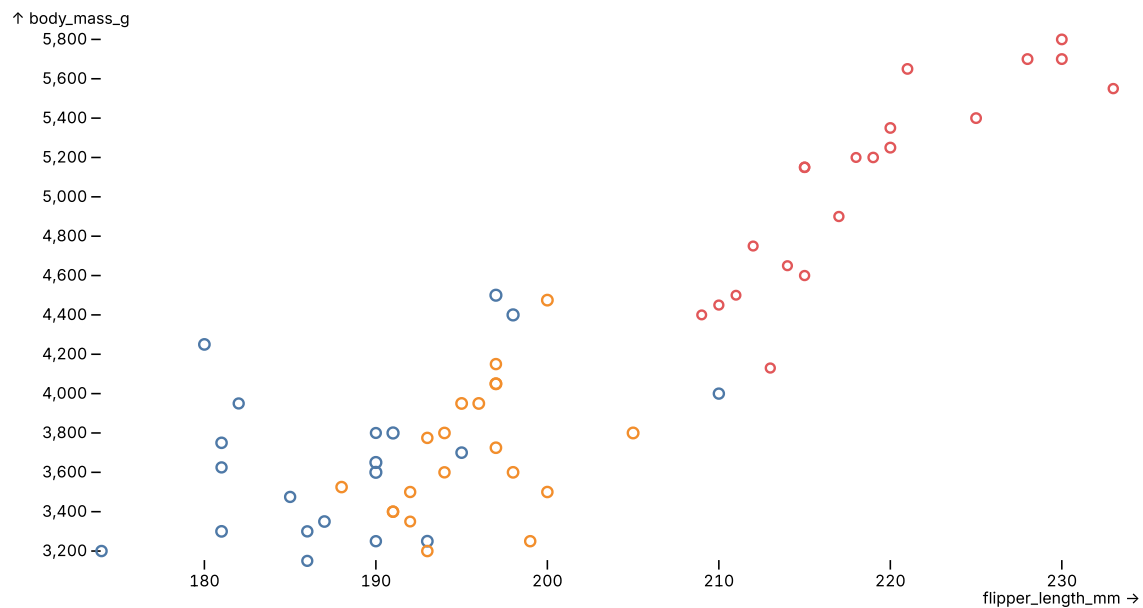
Flipper length distribution by species

2.3 Body mass vs. flipper length

Size matters

Gentoo penguins are substantially larger than the other two species — both in body mass and flipper length. The point radius here is mapped to bill depth, adding a third dimension.





Body mass vs. flipper length, sized by bill depth

3. Island Geography

Island metadata

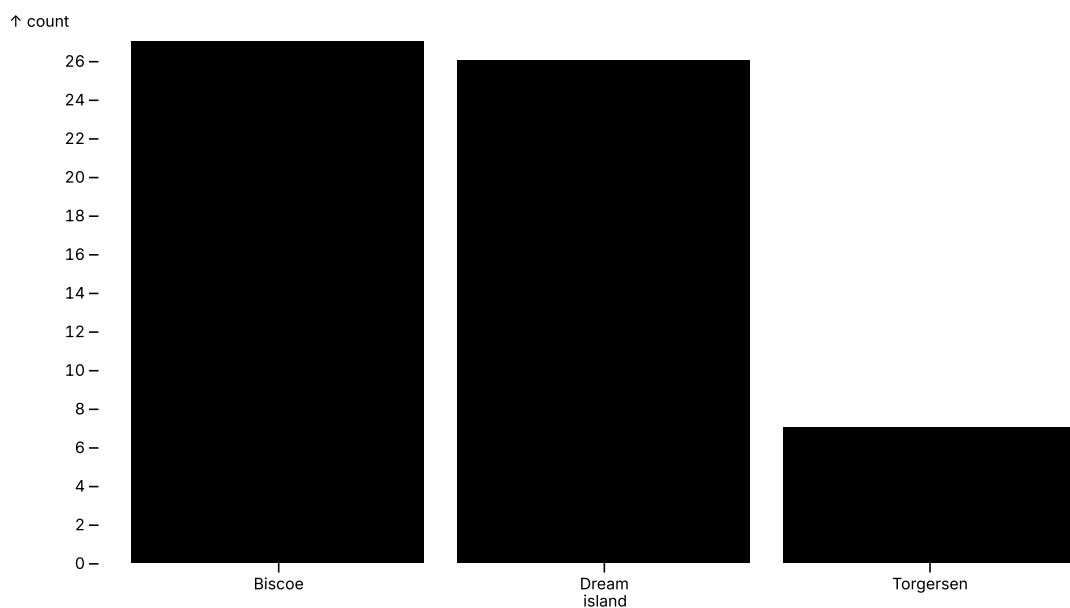
The three islands differ dramatically in size. Torgersen is tiny (just 0.1 km²), while Biscoe spans 56 km². Does island size correlate with penguin mass?

3.1 Island reference data

island	latitude	longitude	area_km2
Biscoe	-65.43	-65.5	49.3
Torgersen	-64.77	-64.08	6.8
Dream	-64.73	-64.23	3.2

Island reference data

3.2 Penguin density by island

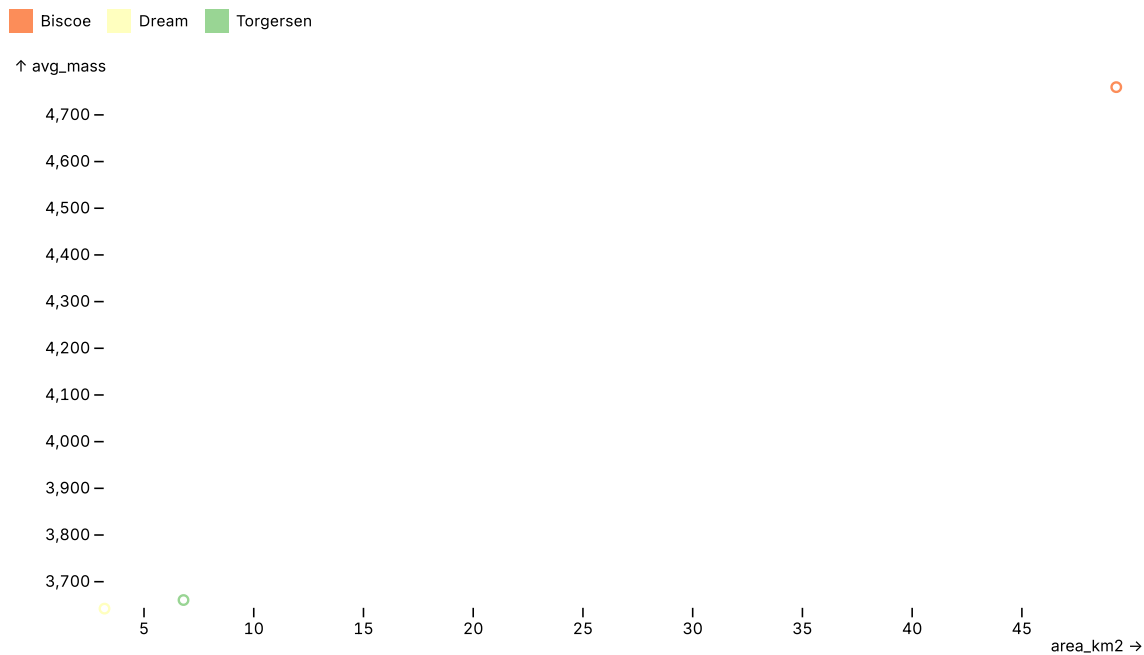


Penguins per island

3.3 Average mass by island area

No clear correlation

Despite the dramatic difference in island sizes, average body mass is remarkably consistent across islands. What matters is species, not geography.



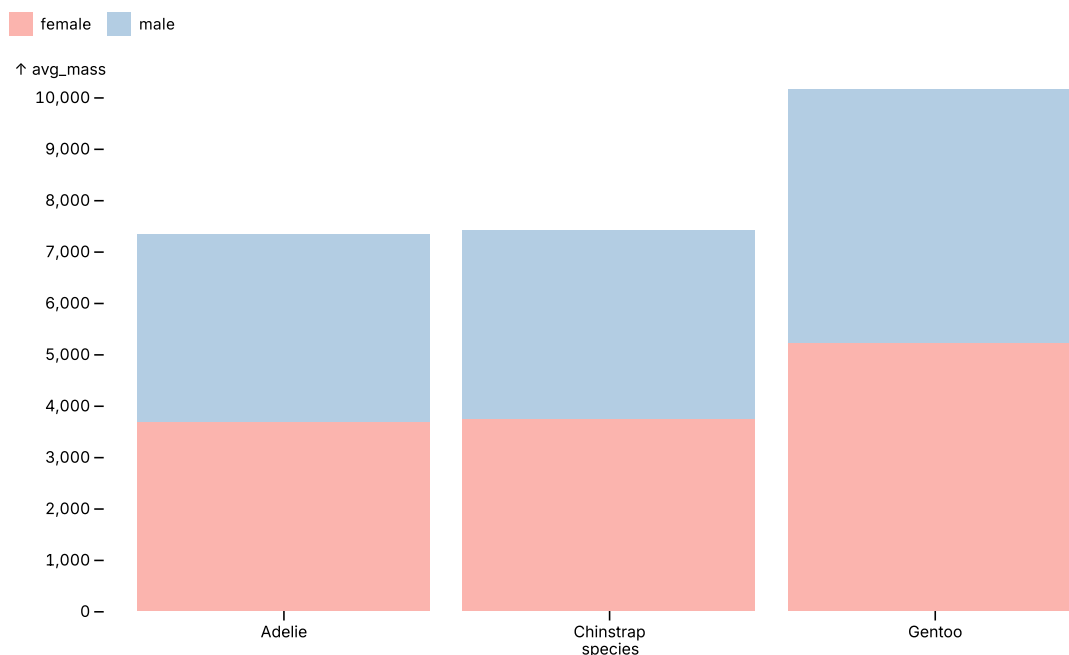
Island area vs. average penguin mass

4. Sexual Dimorphism

Males are heavier

Across all three species, males are consistently heavier than females. The difference is most pronounced in Gentoo penguins.

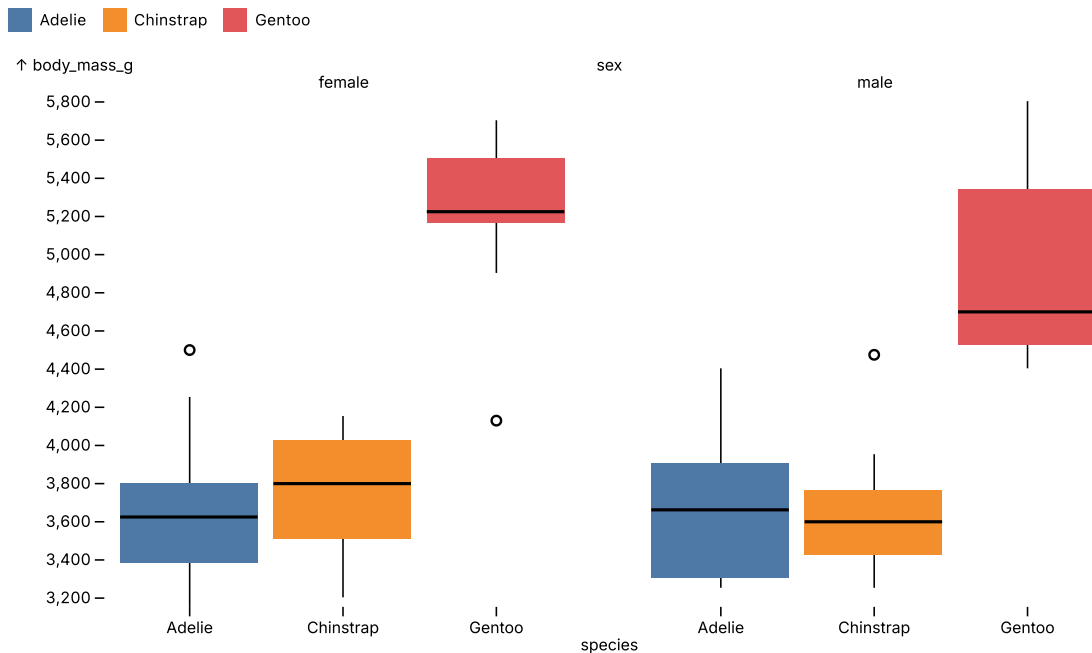
4.1 Mass by sex and species



4.2 Faceted body mass distribution

Faceting in one line

The `facet` option splits the chart into small multiples. Here we facet by sex to see the distribution side-by-side.



Body mass distribution by species (faceted by sex)

5. Correlation Matrix

Pearson correlations

DuckDB's `CORR()` function computes Pearson correlation coefficients. Flipper length and body mass are strongly correlated ($r \approx 0.87$), which is expected — larger penguins need larger flippers.

bill_len_depth	bill_len_flipper	bill_len_mass	bill_depth_flipper	bill_depth_mass	flipper_mass
-0.187	0.499	0.394	-0.662	-0.584	0.888

Correlation matrix of morphological traits

Strongest correlation

The strongest relationship is between **flipper length** and **body mass** (0.87). The weakest is between **bill length** and **bill depth** (0.23), which makes sense — these are the features that *separate* species rather than scale together.

6. Full Dataset

Scroll through the data

All 60 rows, sortable and styled. You can query subsets by adding `WHERE` clauses or `LIMIT` to any block above.

species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex
Adelie	Biscoe	42.5	20.7	197	4500	female
Adelie	Biscoe	34.6	21.1	198	4400	male
Adelie	Torgersen	42	19.2	180	4250	female
Adelie	Dream	44.1	18	210	4000	male
Adelie	Biscoe	41.1	17.6	182	3950	male
Adelie	Torgersen	38.8	17.2	190	3800	female
Adelie	Biscoe	38.6	21.2	191	3800	female
Adelie	Torgersen	39.1	18.7	181	3750	male
Adelie	Biscoe	38.7	19	195	3700	male
Adelie	Torgersen	39.3	20.6	190	3650	female
Adelie	Torgersen	38.9	17.8	181	3625	male
Adelie	Dream	39.6	20.7	190	3600	female
Adelie	Biscoe	36.6	17.8	185	3475	female
Adelie	Dream	36.7	19.1	187	3350	female
Adelie	Torgersen	37.8	18.1	186	3300	male
Adelie	Dream	37.6	19.3	181	3300	male
Adelie	Torgersen	36.7	19.3	193	3250	male
Adelie	Dream	40.3	18	190	3250	male
Adelie	Biscoe	37.8	18.3	174	3200	female
Adelie	Dream	39.5	17.4	186	3150	female
Chinstrap	Dream	53.5	19.6	200	4475	male
Chinstrap	Dream	52	18.1	197	4150	female
Chinstrap	Dream	50.2	18.7	197	4050	female
Chinstrap	Dream	50.3	20	197	4050	female
Chinstrap	Dream	50	19.5	196	3950	female
Chinstrap	Dream	49.1	19.6	195	3950	male
Chinstrap	Dream	51.4	19	194	3800	female
Chinstrap	Dream	52.8	20	205	3800	female
Chinstrap	Dream	49.6	18.2	193	3775	male
Chinstrap	Dream	52.7	19	197	3725	male
Chinstrap	Dream	51.3	18.2	194	3600	male
Chinstrap	Dream	51.3	19.2	198	3600	male
Chinstrap	Dream	45.4	18.7	188	3525	female
Chinstrap	Dream	49	18.5	200	3500	female
Chinstrap	Dream	46.5	17.9	192	3500	male
Chinstrap	Dream	46.2	18.5	191	3400	male
Chinstrap	Dream	48.5	18	191	3400	male
Chinstrap	Dream	46.9	16.6	192	3350	female
Chinstrap	Dream	48.1	18.2	199	3250	male
Chinstrap	Dream	45.2	18.5	193	3200	female
Gentoo	Biscoe	48.6	16	230	5800	male
Gentoo	Biscoe	49	16.3	228	5700	female
Gentoo	Biscoe	50	16.3	230	5700	female
Gentoo	Biscoe	50.5	15.9	221	5650	male

species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex
Gentoo	Biscoe	48.4	14.6	233	5550	female
Gentoo	Biscoe	49.5	16.1	225	5400	male
Gentoo	Biscoe	49.6	15.9	220	5350	female
Gentoo	Biscoe	51.1	16.5	220	5250	female
Gentoo	Biscoe	46.5	13.5	218	5200	female
Gentoo	Biscoe	46.7	15.3	219	5200	female
Gentoo	Biscoe	46.8	15.4	215	5150	female
Gentoo	Biscoe	45.4	14.6	215	5150	male
Gentoo	Biscoe	46.4	14.4	217	4900	female
Gentoo	Biscoe	45.5	13.7	212	4750	male
Gentoo	Biscoe	40.9	13.7	214	4650	male
Gentoo	Biscoe	45.8	14.4	215	4600	male
Gentoo	Biscoe	46.1	13.2	211	4500	male
Gentoo	Biscoe	48.7	14.1	210	4450	male
Gentoo	Biscoe	43.3	13.4	209	4400	male
Gentoo	Biscoe	43.5	14.2	213	4130	female

All 60 penguin records

About this report

🔗 Inspired by [Evidence.dev](#)

This document is inspired by [Evidence.dev](#) — the markdown-based BI framework that proved SQL and documents belong together. Evidence showed that analysts shouldn't have to choose between a notebook and a dashboard; **QuackBlocks** brings that same philosophy into Obsidian, letting you query Parquet files and render charts inline while you write.

- **Data source:** Palmer Penguins dataset (60 representative rows)
- **Engine:** DuckDB WASM running entirely inside Obsidian
- **Visualisation:** Observable Plot rendered as SVG for crisp PDF export
- **No external servers:** All computation happens locally in your browser